

Project Design Phase Solution Architecture

Date	29 July 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Solution Architecture:

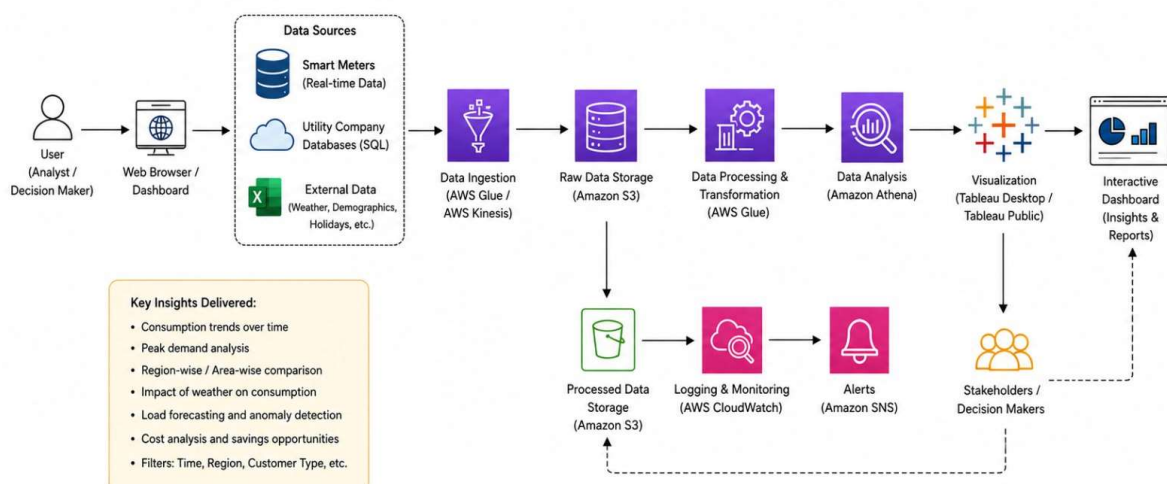
Solution architecture for **"Plugging into the Future: An Exploration of Electricity Consumption Patterns"** is the process of designing an effective data analytics solution that transforms raw electricity consumption data into meaningful insights through interactive visualizations. It bridges the gap between complex electricity datasets and informed decision-making by using Tableau dashboards.

- **Analyze** electricity consumption data to identify trends, seasonal variations, regional differences, and the impact of significant events such as the COVID-19 pandemic.
- **Present** interactive dashboards, charts, maps, and key performance indicators (KPIs) that enable stakeholders to easily understand electricity consumption patterns.
- **Define** the data flow, visualization requirements, dashboard components, filters, and analytical features required for effective electricity consumption analysis.
- **Provide** a scalable and maintainable analytics solution that allows users to monitor electricity consumption, compare state-wise performance, generate reports, and support data-driven energy planning and policy decisions.

Example - Solution Architecture Diagram:

Example - Solution Architecture Diagram:

Project: "Plugging into the future: An exploration of electricity consumption patterns."



Reference: <https://aws.amazon.com/architecture/solutions/>

Figure 1: Architecture and data flow of the voice patient diary sample application